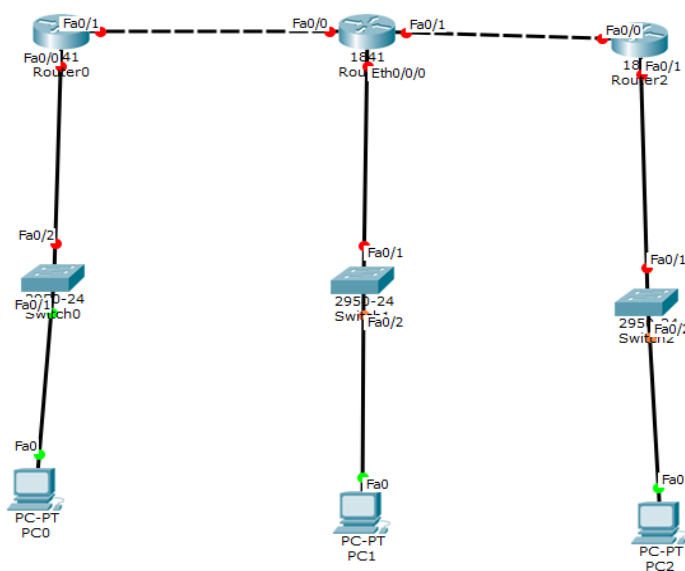
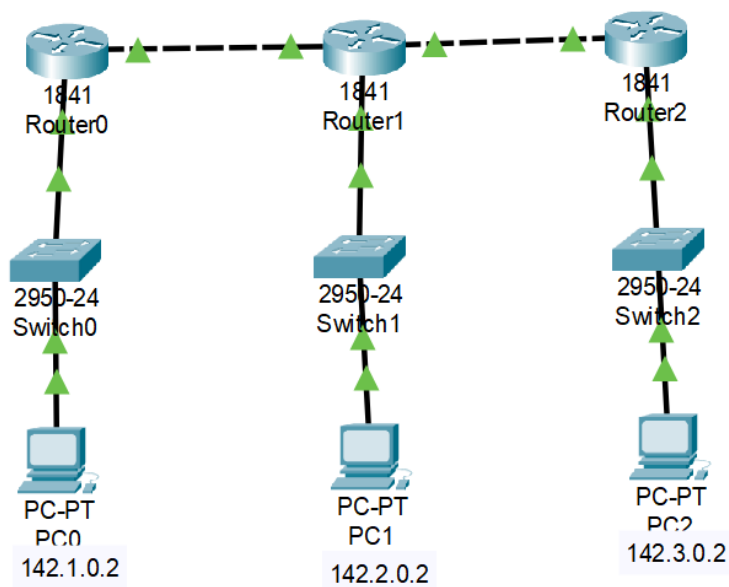


Date:

EXPERIMENT NO. 6**AIM: Understand & identify Packet(L3) & frame(L2) content detail.**

Topology for the consideration
Student Topology with IP address and MAC address as label



Steps:

1. Create topology in Realtime mode
2. Configure IP address
3. Configure Dynamic Routing in Each routers
4. Ping from PC0 to rest other PCs and all interface of routers and fill success table
5. Go to Simulation mode
6. Prepare MAC and IP address Table.
7. Prepare ARP table for all PCs
8. Prepare Routing tables for Router0, Router1 and Router2
9. Prepare ARP tables for Router0, Router1 and Router2
10. Prepare MAC table of all switches
11. In simulation mode follow instructions as given in exercise and write answer of questions.

Destination machine	Destination IP address	Command	Success/Fail
FE0/0 of Router0	142.1.0.1	Ping 142.1.0.1	Success
FE0/1 of Router0	142.4.0.2	Ping 142.4.0.2	Success
FE0/0 of Router1	142.2.0.1	Ping 142.2.0.1	Success
FE0/1 of Router1	142.4.0.3	Ping 142.4.0.3	Success
FE0/0/0 of Router1	142.5.0.2	Ping 142.5.0.2	Success
PC1	142.2.0.2	Ping 142.2.0.2	Success
FE0/0 of Router2	142.3.0.1	Ping 142.3.0.1	Success
FE0/1 of Router2	142.5.0.3	Ping 142.5.0.3	Success
PC2	142.3.0.2	Ping 142.3.0.2	Success

Ping Success table

Ping from PC0 to PC2

Computer/Router Interface	MAC address	IP address
PC0	0060.2FD6.6ECD	142.1.0.2
Router0 FE0/0	0005.5EA8.4E01	142.1.0.1
Router0 FE0/1	0005.5EA8.4E02	142.4.0.2
Router1 FE0/0	0060.4765.0E01	142.2.0.1
Router1 FE0/1	0060.4765.0E02	142.4.0.3
Router1 ETH0/0/0	000C.850B.8270	142.5.0.2
PC1	000A.413B.4EED	142.2.0.2

Router2 FE0/0	0060.5C2C.2801	142.3.0.1
Router2 FE0/1	0060.5C2C.2802	142.5.0.3
PC2	0001.966D.6AA9	142.3.0.2

MAC and IP address Table

IP Address	MAC Address	Interface
142.1.0.1	0005.5EA8.4E01	FastEthernet0

ARP Table for PC0

IP Address	MAC Address	Interface
142.2.0.1	0060.4765.0E01	FastEthernet0

ARP Table for PC1

IP Address	MAC Address	Interface
142.3.0.1	0060.5C2C.2801	FastEthernet0

ARP Table for PC2

Type	Network	Port	Next Hop ID	Metric
C	142.1.0.0/16	FastEthernet0/0	--	0/0
R	142.2.0.0/16	FastEthernet0/1	142.4.0.3	120/1
R	142.3.0.0/16	FastEthernet0/1	142.4.0.3	120/2
C	142.4.0.0/16	FastEthernet0/1	--	0/0
R	142.5.0.0/16	FastEthernet0/1	142.4.0.3	120/1

Routing table for Router0

Type	Network	Port	Next Hop ID	Metric
R	142.1.0.0/16	FastEthernet0/1	142.4.0.2	120/1
C	142.2.0.0/16	FastEthernet0/0	--	0/0
R	142.3.0.0/16	Ethernet0/0/0	142.5.0.3	120/1
C	142.4.0.0/16	FastEthernet0/1	--	0/0
C	142.5.0.0/16	Ethernet0/0/0	--	0/0

Routing table for Router1

Type	Network	Port	Next Hop ID	Metric
R	142.1.0.0/16	FastEthernet0/1	142.5.0.2	120/2
R	142.2.0.0/16	FastEthernet0/1	142.5.0.2	120/1
C	142.3.0.0/16	FastEthernet0/0	--	0/0
R	142.4.0.0/16	FastEthernet0/1	142.5.0.2	120/1
C	142.5.0.0/16	FastEthernet0/0	--	0/0

Routing table for Router2

IP Address	MAC Address	Interface
142.1.0.1	0005.5EA8.4E01	FastEthernet0/0
142.1.0.2	0060.2FD6.6ECD	FastEthernet0/0
142.4.0.2	0005.5EA8.4E02	FastEthernet0/1
142.4.0.3	0060.4765.0E02	FastEthernet0/1

ARP Table for Router0

IP Address	MAC Address	Interface
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142.2.0.1	0060.4765.0E01	FastEthernet0/0
142.2.0.2	000A.413B.4EED	FastEthernet0/0
142.4.0.2	0005.5EA8.4E02	FastEthernet0/1
142.4.0.3	0060.4765.0E02	FastEthernet0/1
142.5.0.2	000C.850B.8270	Ethernet0/0/0
142.5.0.3	0060.5C2C.2802	Ethernet0/0/0

IP Address	MAC Address	Interface
142.3.0.1	0060.5C2C.2801	FastEthernet0/0
142.3.0.2	0001.966D.6AA9	FastEthernet0/0
142.5.0.2	000C.850B.8270	FastEthernet0/1
142.5.0.3	0060.5C2C.2802	FastEthernet0/1

ARP Table for Router2

VLAN ID	MAC Address	port
1	0005.5EA8.4E01	FastEthernet0/2

Switching table for Switch0

VLAN ID	MAC Address	port
1	0060.4765.0E01	FastEthernet0/2

Switching table for Switch1

VLAN ID	MAC Address	port
1	0060.5C2C.2801	FastEthernet0/2

Switching table for Switch2

In simulation mode**Exercise-1:** Ping from PC0 to FE0/0 of Router0

click on capture forward once so packet goes to switch, Inspect& write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	0005.5EA8.4E01
Destination MAC address	0000.0000.0000	0060.2FD6.6ECD
Source IP Address	142.1.0.2	142.1.0.1
Destination IP address	142.1.0.1	142.1.0.2

Question: What decision will be taken by switch?

Exercise-2: Ping from PC0 to FE0/1 of Router0

Click on capture forward once so packet goes to switch, Inspect& write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	0005.5EA8.4E01
Destination MAC address	0005.5EA8.4E01	0060.2FD6.6ECD
Source IP Address	142.1.0.2	142.4.0.2
Destination IP address	142.4.0.2	142.1.0.2

Question: Is there any difference between table content of exercise-1 and 2? Why?

Exercise-3: Ping from PC0 to FE0/0 of Router1

Click on capture forward once so packet goes to switch, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	0060.2FD6.6ECD
Destination MAC address	0005.5EA8.4E01	0005.5EA8.4E01
Source IP Address	142.1.0.2	142.1.0.2
Destination IP address	142.2.0.1	142.2.0.1

Click on capture forward once so packet goes to Router0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0005.5EA8.4E01	0005.5EA8.4E01
Destination MAC address	0060.2FD6.6ECD	0060.2FD6.6ECD
Source IP Address	142.1.0.1	142.1.0.1
Destination IP address	142.1.0.2	142.1.0.2

Question: What decision will be taken by Router0?

Question: Is Inbound and outbound PDU detail remain same? If not why?

Exercise-4: Ping form PC0 to PC1 (For even roll number of student PC0 to PC2)

Click on capture forward so packet goes to switch0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	0005.5EA8.4E01
Destination MAC address	0005.5EA8.4E01	0060.2FD6.6ECD
Source IP Address	142.1.0.2	142.3.0.2
Destination IP address	142.3.0.2	142.1.0.2

Click on capture forward so packet goes to Router0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	0005.5EA8.4E01
Destination MAC address	0005.5EA8.4E01	0060.4765.0E02
Source IP Address	142.1.0.2	142.1.0.2
Destination IP address	142.3.0.2	142.3.0.2

Click on capture forward so packet goes to Router2, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.5C2C.2801	000C.850B.8270
Destination MAC address	0001.966D.6AA9	0060.5C2C.2802
Source IP Address	142.1.0.2	142.1.0.2
Destination IP address	142.3.0.2	142.3.0.2

Click on capture forward so packet goes to switch2, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.5C2C.2801	0060.5C2C.2801
Destination MAC address	0001.966D.6AA9	0001.966D.6AA9
Source IP Address	142.1.0.2	142.1.0.2
Destination IP address	142.3.0.2	142.3.0.2

Click on capture forward so packet goes to PC2, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.5C2C.2801	0060.5C2C.2801
Destination MAC address	0001.966D.6AA9	0001.966D.6AA9
Source IP Address	142.1.0.2	142.3.0.2
Destination IP address	142.3.0.2	142.1.0.2

Click on capture forward so packet goes to switch2, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0001.966D.6AA9	0001.966D.6AA9
Destination MAC address	0060.5C2C.2801	0060.5C2C.2801
Source IP Address	142.3.0.2	142.3.0.2
Destination IP address	142.1.0.2	142.1.0.2

Click on capture forward so packet goes to Router2, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0001.966D.6AA9	0060.5C2C.2802
Destination MAC address	0060.5C2C.2801	000C.850B.8270
Source IP Address	142.3.0.2	142.3.0.2
Destination IP address	142.1.0.2	142.1.0.2

Click on capture forward so packet goes to router0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.4765.0E02	0005.5EA8.4E01
Destination MAC address	0005.5EA8.4E02	0060.2FD6.6ECD
Source IP Address	142.3.0.2	142.3.0.2
Destination IP address	142.1.0.2	142.1.0.2

Click on capture forward so packet goes to switch0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0005.5EA8.4E01	0005.5EA8.4E01
Destination MAC address	0060.2FD6.6ECD	0060.2FD6.6ECD
Source IP Address	142.3.0.2	142.3.0.2
Destination IP address	142.1.0.2	142.1.0.2

Click on capture forward so packet goes to PC0, Inspect & write Inbound and Outbound PDU and fill following table

	In Bound	Out Bound
Source MAC Address	0060.2FD6.6ECD	--
Destination MAC address	0005.5EA8.4E01	--
Source IP Address	142.1.0.2	--
Destination IP address	142.3.0.2	--

Observe/inspect values of above tables and answer following questions.

Question: Is Source IP and Destination IP remains same for one way of data transmission?

Justify: In reverse direction, source IP and destination IP address gets changed.

CONCLUSIONS: